

Group 16 Project Proposal:

Oil & Gas Production Forecasting

Team Members:

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I. Introduction

Oil production can change over time because of factors such as well pressure, temperature, choke settings, operating time, and reservoir changes. Accurate production forecasting is an important business problem because it can help operators anticipate production trends and support operational and planning decisions.

We will use the Equinor Volve Production Dataset, which contains real-world production data from the Volve oil field in the North Sea. We will focus on the daily production dataset, which contains 15,634 observations and 24 columns. The dataset includes production and operating variables such as operating hours, downhole pressure and temperature, tubing pressure, choke size, wellhead pressure and temperature, and daily oil, gas, and water production.

The main goal of this project is to use historical production and operating data to predict future oil production and compare traditional machine learning methods with deep learning methods. Our research question is: Can historical well production and operating conditions predict future oil production? To answer this question, we will compare a traditional statistical baseline, a traditional machine learning model, and a deep learning model: Linear Regression, XGBoost, and LSTM. By comparing Linear Regression, XGBoost, CNN, and LSTM, we will examine whether traditional machine learning or deep learning is more effective for this dataset. We also want to identify which production and operating variables have the strongest relationships with future oil production.

The project demonstrates how data science can be applied to a real-world oil and gas problem where better forecasting can support production planning and operational decision-making.

II. Motivation

For this project, we chose oil production forecasting because we wanted to address a real-world business problem using data science. Oil production forecasting is an important problem in the

oil and gas industry because production levels can change over time as operating conditions and well behavior change. Being able to estimate future production from historical information can provide useful information for production planning and operational decision-making.

From a data science perspective, this project also provides an opportunity to investigate how different types of machine learning models handle a real-world time-dependent dataset. Rather than assuming that a more complex deep learning model will always perform better, we want to compare models with different levels of complexity and determine whether the additional complexity of CNN and LSTM provides a meaningful improvement over simpler approaches.

III. Evaluation

We will consider the project successful if we can develop models that produce reasonable predictions of future daily oil production and determine how the three approaches compare on previously unseen data. We will evaluate and compare the models using the following regression metrics:

Mean Absolute Error (MAE): Measures the average absolute difference between the predicted and actual oil production values. A lower MAE means the model's predictions are closer to the actual values.

Root Mean Squared Error (RMSE): Measures the difference between predicted and actual production while giving more weight to larger errors. A lower RMSE indicates better prediction performance.

R² Score: Measures how much of the variation in oil production can be explained by the model. An R² value closer to 1 indicates that the model explains more of the variation in oil production.

Since this is time-based production data, we will split the data based on time rather than randomly. Earlier observations will be used for training, while later observations will be used for testing. This helps simulate the real-world situation where historical data is used to predict future production.

IV. Resources

Dataset: Christian, Albert Lamy. “Volve Production Data.” *Kaggle*, 31 May 2022, www.kaggle.com/datasets/lamyalbert/volve-production-data.

Features and Target variables:

The dataset includes both daily and monthly records from multiple wells. However, we will only use the daily dataset because it provides more production-related metrics and higher temporal resolution for analyzing relationships between operating conditions and production. It provides information such as well pressure, temperature, choke size, operating hours, and the amount of

oil, gas, and water produced or injected. These variables are used as **features** (inputs) by the machine learning and deep learning models to identify patterns related to oil production. The **target variable** for this project is BORE_OIL_VOL, which represents the daily oil production volume of a well.

Train and Test data:

Because this is a forecasting problem, the data will be divided chronologically, with earlier observations used for training and later observations used for testing. This will simulate the real-world situation of using historical data to predict future production and will help prevent future information from entering the training process

The table Daily Production Data has 15,634 daily observations and 24 columns.

Purpose	Columns
Target Variable	BORE_OIL_VOL
Operating Conditions	ON_STREAM_HRS, AVG_DOWNHOLE_PRESSURE, AVG_DOWNHOLE_TEMPERATURE, AVG_DP_TUBING, AVG_ANNULUS_PRESS, AVG_CHOKE_SIZE_P, AVG_WHP_P, AVG_WHT_P, DP_CHOKE_SIZE
Time Information	DATEPRD
Historical Production Features	BORE_OIL_VOL_lag1, BORE_OIL_VOL_lag2, BORE_OIL_VOL_lag7, 3-day/7-day rolling averages
Well Identification	WELL_BORE_CODE
Well Metadata (excluded)	NPD_WELL_BORE_CODE, NPD_WELL_BORE_NAME, NPD_FIELD_CODE, NPD_FIELD_NAME, NPD_FACILITY_CODE, NPD_FACILITY_NAME, AVG_CHOKE_UOM
Well Classification Variables (excluded)	FLOW_KIND, WELL_TYPE
Other Production/Injection Outputs (excluded)	BORE_GAS_VOL, BORE_WAT_VOL, BORE_WI_VOL

The project will be implemented in Python using standard data science and machine learning tools. We will use libraries such as pandas and NumPy for data processing, scikit-learn for Linear Regression and evaluation metrics, XGBoost for gradient-boosted tree regression, and TensorFlow/Keras or another appropriate deep learning framework for the CNN and LSTM

model. Matplotlib and other visualization tools will be used to examine production trends and model results.

V. Team contribution

Both analytical and technical work will be split evenly between 3 team members. Every member will be involved in writing the project proposal, final project report, presentation, and poster.

Team Member	Responsibilities
Thy	1. Explore the dataset 2. Clean and preprocess the data 3. Feature engineering 5. Train Linear Regression
Buu	4. Create training and testing sets 6. Train XGBoost 7. Train 1D CNN
An	8. Train LSTM 9. Evaluate the models 10. Analyze the results

VI. Potential Challenges

One challenge is missing data because some measurements may not be available for every well and every day. We will examine the amount of missing data before deciding how to handle it.

Another challenge is that different wells may have different production behaviors. A model that works well for one well may not perform equally well for another well.

We also need to prevent data leakage. Since the goal is to predict future production, information from future dates should not be used to predict earlier dates. Therefore, the training and testing data will be separated chronologically.

The LSTM and CNN models may also require additional preprocessing and training time compared with Linear Regression and XGBoost. We will adjust the model structure and parameters if necessary while keeping the comparison between models fair.